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HEAD-OF-THE-LINE BLOCKING FREE PROTOCOL FOR MULTIPLEXED CONNECTIONS

Abstract

A sender and a receiver exchange packets of messages from multiple flows. Each packet includes a flow and a packet sequence number (PSN). An initial packet indicates a total number of packets in the message. The receiver receives a first packet of a first message of a first flow and discards the first packet if the first message cannot be processed. The receiver receives a second packet and discards the second packet if the second packet is of the first message, belongs to the first flow, or is of a different message and does not belong to the first flow, and a PSN of the second received packet is not equal to a sum of a PSN of the first packet, the total number of packets of the first message, and the total number of packets of any message discarded after discarding the first packet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/EP2022/075681, filed on Sep. 15, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD
[0002] The present disclosure relates to a sender and a receiver for respectively sending and receiving packets of messages from multiple flows multiplexed onto a connection between the receiver and the sender. For instance, the connection between the receiver and the sender can be a remote direct memory access (RDMA) connection. The disclosure provides a protocol for sending and receiving packets on such a multiplexed connection, wherein the protocol is head-of-the-line (HOL) blocking free.
BACKGROUND
[0003] RDMA is a technology that is widely used in modern datacenters and computer clusters for low-latency and high-bandwidth networking. RDMA is based on offloading memory operations from a core processing unit (CPU) to a RDMA network interface card (RNIC), which directly accesses a memory. This offload saves CPU time, so that the CPU is free to perform other tasks. The RDMA protocol introduces a real zero copy transmission, wherein user buffers are copied directly to the network without any copies on the way.
[0004] The software-layer of the RDMA protocol uses so-called “verbs” to perform various RDMA operations, which are then transformed to work-requests written to queues of the RNIC. Such a work-request is called a work queue element (WQE). RDMA operations are either tagged (one-side WRITE/READ/ATOMIC) or untagged (two-sided SEND/RECEIVE). RDMA operations may have a maximum data length of 2 GB, and are sent over the underlining fabric in multiple packets.
[0005] RDMA peers are communicating over queue-pairs (QPs) offering various transport services. The common QP types are: reliable connection (RC); reliable datagram (RD), extended reliable connection (XRC); unreliable datagram (UD); and unreliable connection (UC).
[0006] Different companies also have proprietary QP types, which allow multiplexing different destinations or applications for a higher reliable service scale. For instance, scalable reliable datagram (SRD) allows application multiplexing as a reliable datagram service, and dynamic connection (DC) or dynamically connected transport (DCT) allows destination multiplexing as a reliable connection service per destination.
SUMMARY
[0007] Multiplexed connections—such as RDMA RD QP type—may suffer from a limitation that reduces the performance drastically. In particular, the multiplexed connections may suffer from so-called HOL blocking. HOL blocking occurs when packets are held up in a queue by a blocked packet. For example, HOL blocking occurs, if there is a single queue of packets waiting to be transmitted, and the packet at the head of the queue (line) cannot move forward due to a congestion or due to an inability to process that packet, even if other packets behind that packet could move forward. HOL blocking makes, for example, the above-mentioned RDMA RD QP type a service, which is not implemented in the industry.
[0008] Therefore, embodiments of the present disclosure provides an improved transport protocol for multiplexing multiple logical connections into a single physical connection, i.e., for a

multiplexed connection. In accordance with another embodiment the transport protocol is reliable and avoids the HOL blocking issue. In accordance with another embodiment, an order is maintained for each of multiple flows multiplexed onto the multiplexed connection.

[0009] In this context, “reliable transport” means that a message, which includes one or more packets, is guaranteed to be exchanged exactly once. Multiplexing multiple logical connections means that all messages to a specific destination node are managed and exchanged using a single connection. Multiple destinations means that each message includes its own destination information, unlike for instance for RDMA RC operations, in which a QP context holds such information and all messages go to the same destination.

[0010] A first aspect of this disclosure provides a receiver for receiving packets of messages from multiple flows multiplexed onto a connection between the receiver and a sender; wherein each packet indicates at least the flow to which it belongs and its packet sequence number (PSN), and wherein an initial packet of each message indicates at least a total number of packets of that message. The receiver is configured to receive, from the sender, a first received packet that is an initial packet of a first message of a first flow; discard the first received packet, if the first message cannot be processed by the receiver; receive, from the sender, a second received packet; and discard the second received packet under each one of the following conditions; the second received packet is of the first message; the second received packet is of a second message and belongs to the first flow; the second received packet is of a third message and does not belong to the first flow, and the PSN of the second received packet is not equal to the sum of the PSN of the first received packet plus the total number of packets of the first message plus the total number of packets of any message that belongs to any flow and was discarded after discarding the first received packet.

[0011] Due to the conditions, under which the second received packet is disregarded by the receiver of the first aspect, the sending of the packets on the (multiplexed) connection between the sender and the receiver, is HOL blocking free. The first received packet, even though it cannot be processed, and the second received packet, do not hold up further packets that could move forward. Thus, an improved and reliable transport protocol for the multiplexed connection is provided.

[0012] In an implementation form of the first aspect, the receiver is further configured to maintain flow status information indicating one or more blocked flows of the multiple flows; and discard the second received packet under the condition that the second received packet belongs to a second flow that is a blocked flow according to the flow status information.

[0013] The flow status information allows the receiver to track blocked flows, and also to provide this information to the sender. For example, a list containing the first discarded message per blocked flow can be provided to the sender. The flow status information can then be derived by the sender from this list, optionally based on its local data structures. This is beneficial for the HOL blocking free protocol.

[0014] In an implementation form of the first aspect, if the first message cannot be processed by the receiver, the receiver is further configured to modify the flow status information to indicate that the first flow is a blocked flow.

[0015] Thus, the receiver can update the information indicating blocked flows, and the updated information can be shared with the sender.

[0016] In an implementation form of the first aspect, the receiver is further configured to maintain PSN information which indicates the PSN of an expected packet, which is expected to be received next from the sender; discard the second received packet under the condition that the PSN of the second received packet is not equal to the PSN of the expected packet as indicated by the PSN information; and not discard the second received packet under the condition that the second received packet belongs to a third flow that has a status, according to the flow status information, which indicates that the third flow is not blocked, and if the PSN of the second received packet is equal to the PSN of the expected packet as indicated by the PSN information, and if the third message can be processed by the receiver.

[0017] This allows an order of messages for each of multiple flows multiplexed onto the multiplexed connection to be maintained.

[0018] In an implementation form of the first aspect, the receiver is further configured to, if the first message cannot be processed by the receiver, increase the PSN of the expected packet by the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet.

[0019] Generally, the receiver is configured to increase the PSN of the expected packet for every received initial packet of any message that belongs to a blocked flow. This supports the HOL blocking free protocol.

[0020] In an implementation form of the first aspect, each message is associated with a message sequence number (MSN), and the receiver is further configured to maintain MSN information which indicates the MSN of an expected message, which is expected to be processed next; and increase the MSN indicated by the MSN information by one, if the first message cannot be processed.

[0021] In an implementation form of the first aspect, the receiver is further configured to increase the MSN indicated by the MSN information by one for each second message that belongs to the first flow and was discarded after discarding the first received packet.

[0022] The tracking and updating of the MSN information at the receiver supports the HOL blocking free protocol, and supports maintaining an order of all flows.

[0023] In an implementation form of the first aspect, the receiver is further configured to increase the MSN indicated by the MSN information by one, if the second received packet is the last packet of a message and belongs to the third flow.

[0024] In an implementation form of the first aspect, the receiver is further configured to send a first not-acknowledge (NACK) packet to the sender, if the first message cannot be processed by the receiver; wherein the first NACK packet includes the PSN information and an indication that the first message cannot be processed by the receiver.

[0025] Thus, the sender can accordingly adapt, to avoid HOL blocking.

[0026] In an implementation form of the first aspect, the receiver is further configured to send a second NACK packet to the sender, if the second received packet is discarded and if the second received packet is of the second message or of the third message; wherein the second NACK packet includes the PSN information and an indication that the first message cannot be processed by the receiver.

[0027] In an implementation form of the first aspect, the first NACK packet and the second NACK packet include the MSN of the first message that cannot be processed by the receiver.

[0028] In an implementation form of the first aspect, the first NACK packet and the second NACK packet include an MSN list comprising, for each blocked flow, the MSN of the first blocked message of said blocked flow; or the first NACK packet and the second NACK packet include a PSN list comprising, for each blocked flow, the PSN of the initial packet of the first blocked message of said blocked flow.

[0029] In an implementation form of the first aspect, the receiver is further configured to send an acknowledge (ACK) packet to the sender, if the second received packet is not discarded; wherein the ACK message includes the PSN of the second received packet, the MSN of the message the second received packet belongs to, and the MSN list comprising, for each blocked flow, the MSN of the first blocked message of said blocked flow.

[0030] In an implementation form of the first aspect, the receiver is further configured to receive, from the sender, a third received packet of a fourth message; discard the third received packet in case of a predetermined failure condition and modify the flow status information of the flow to which the fourth message belongs to indicate that this flow is a blocked flow; send a third NACK packet to sender, if the third received packet is discarded; and hold the PSN of the third received packet and/or the MSN of the fourth message.

[0031] The failure condition may be a page fault. The NACK packet may be a RNR message. This disclosure supports blocking of a flow for any packet received (not only the initial packet of a message). This may be achieved by identifying either or both the MSN and the PSN of the received packet that caused the flow to be blocked. The expected PSN may be incremented according to the number of remaining packets in the message, and the MSN may be incremented by 1.
[0032] In an implementation form of the first aspect, the third NACK packet includes the PSN of the third packet and/or the MSN of the fourth message.
[0033] In an implementation form of the first aspect, the connection between the receiver and the sender is a RDMA connection and/or is a RD connection.
[0034] Thus, the reliable transport protocol of this disclosure is compatible with RD RDMA.
[0035] A second aspect of this disclosure provides a sender sending packets of messages on multiple flows multiplexed onto a connection between the sender and a receiver; [0036] wherein each packet indicates at least the flow to which it belongs and its PSN, and an initial packet of each message indicates at least a total number of packets of that message. The sender is configured to send, to the receiver, a first sent packet that is an initial packet of a first message on a first flow; receive, from the receiver, a NACK packet that includes PSN information which indicates a PSN of an expected packet, which is expected to be received next at the receiver, and an indication that the first message cannot be processed by the receiver; abort sending any further packet of the first message; and send, to the receiver, a second sent packet that does not belong to the first flow, wherein the PSN of the second sent packet is equal to the sum of the PSN of the first sent packet plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was sent to the receiver after the first sent packet.
[0037] Due to the aborting of sending packets of the first message, and then sending of the second sent packet, the sending of the packets on the (multiplexed) connection between the sender and the receiver, can be HOL blocking free. Thus, an improved and reliable transport protocol for the multiplexed connection is provided.
[0038] In an implementation form of the second aspect, the NACK packet includes an MSN list comprising, for each blocked flow, the MSN of the first message that cannot be processed by the receiver or is discarded by the receiver, and the sender is further configured to skip sending any first transmission of a packet that belongs to a flow, on which any message that cannot be processed by the receiver is sent.
[0039] In an implementation form of the second aspect, the sender is further configured to maintain send sequence number (SSN) information which indicates a MSN of a next message to be sent; and send the second sent packet, and associate the second message with a MSN as indicated by the SSN information.
[0040] In an implementation form of the second aspect, the sender is further configured to associate the second message with the MSN of the third message, if the third message was planned to be sent before the second message but was skipped.
[0041] A third aspect of this disclosure provides a method for receiving packets of messages from multiple flows multiplexed onto a connection between a receiver and a sender; wherein each packet indicates at least the flow to which it belongs and its packet sequence number, PSN, and wherein an initial packet of each message indicates at least a total number of packets of that message. The method comprises receiving, from the sender, a first received packet that is an initial packet of a first message of a first flow; discarding the first received packet, if the first message cannot be processed by the receiver; receiving, from the sender, a second received packet; and discarding the second received packet under each one of the following conditions: the second received packet is of the first message; the second received packet is of a second message and belongs to the first flow; the second received packet is of a third message and does not belong to the first flow, and the PSN of the second received packet is not equal to the sum of the PSN of the first received packet plus the total number of packets of the first message plus the total number of packets of any

message that belongs to any flow and was discarded after discarding the first received packet.

[0042] The method of the third aspect may have implementation forms, which correspond to the implementation forms of the receiver of the first aspect. The method of the third aspect and its implementation forms achieves the same advantages as the receiver of the first aspect and its corresponding implementation forms.

[0043] A fourth aspect of this disclosure provides a method for sending packets of messages on multiple flows multiplexed onto a connection between a receiver and a sender; wherein each packet indicates at least the flow to which it belongs and its PSN, and an initial packet of each message indicates at least a total number of packets of that message. The method comprises sending, to the receiver, a first sent packet that is an initial packet of a first message on a first flow; receiving, from the receiver a NACK, packet that includes PSN information which indicates a PSN of an expected packet, which is expected to be received next at the receiver, and an indication that the first message cannot be processed by the receiver; aborting sending any further packet of the first message; and sending, to the receiver, a second sent packet that does not belong to the first flow, wherein the PSN of the second sent packet is equal to the sum of the PSN of the first sent packet plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was sent to the receiver after the first sent packet.

[0044] The method of the fourth aspect may have implementation forms, which correspond to the implementation forms of the sender of the second aspect. The method of the fourth aspect and its implementation forms achieves the same advantages as the sender of the second aspect and its corresponding implementation forms.

[0045] A fifths aspect of this disclosure provides a computer program comprising instructions which, when the program is executed by a computer, cause the computer to perform the method according to the third aspect or fourth aspect.

[0046] A sixth aspect of this disclosure provides a non-transitory storage medium storing executable program code which, when executed by a processor, causes the method according to the third or fourth aspect to be performed.

[0047] According to the above, the disclosure provides a reliable transport protocol that is HOL blocking free. The receiver may maintain a database of blocked MSNs and blocked flows. When a flow becomes blocked, the MSN of the message that caused the blocking may be added to the database, and the flow may be marked as blocked. The list of blocked MSNs may be carried in each NACK or ACK packet sent back to the sender. The receiver may set its flow status information, MSN information, and PSN information, such that the next message received in-order from a flow that is not blocked is accepted. When the sender receives a blocked MSN for the first time, it may mark the flow corresponding with the SSN information as blocked, and it may increment the SSN according to the SSN information, as if it was completed normally. The sender may remember for each message (e.g., WQE) corresponding to a specific SSN, whether it was already transmitted and whether it is resumed.

[0048] It has to be noted that all devices, elements, units and means described in the present application could be implemented in the software or hardware elements or any kind of combination thereof. All steps which are performed by the various entities described in the present application as well as the functionalities described to be performed by the various entities are intended to mean that the respective entity is adapted to or configured to perform the respective steps and functionalities. Even if, in the following description of specific embodiments, a specific functionality or step to be performed by external entities is not reflected in the description of a specific detailed element of that entity which performs that specific step or functionality, it should be clear for a skilled person that these methods and functionalities can be implemented in respective software or hardware elements, or any kind of combination thereof.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0049] The above described aspects and implementation forms will be explained in the following description of specific embodiments in relation to the enclosed drawings, in which

[0050] FIG. **1** shows a sender and a receiver according to this disclosure for respectively sending and receiving packets of messages from multiple flows multiplexed on a connection between the sender and the received.

[0051] FIG. **2** shows an example of accepting a message in the presence of a blocked flow at a receiver according to an embodiment of the present disclosure.

[0052] FIG. **3** shows re-transmitting a packet of a blocked flow by a sender according to an embodiment of the present disclosure.

[0053] FIG. **4** shows skipping a message of a blocked flow at a sender according to an embodiment of the present disclosure.

[0054] FIG. **5** shows a method for receiving packets according to an embodiment of the present disclosure.

[0055] FIG. **6** shows a method for sending packets according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0056] FIG. **1** shows a receiver **100** according to this disclosure, and a sender **110** according to this disclosure. The receiver **100** is configured to receive packets of messages from multiple flows multiplexed onto a connection between the receiver **100** and the sender **110**. The sender **110** is accordingly configured to send packets of messages on multiple flows multiplexed onto the connection. The connection between the receiver **100** and the sender **110** may be a RDMA connection and/or may a RD connection, for instance, may be a RDMA RD connection using a RDMA RD QP type. Accordingly, the receiver **100** may be a RDMA receiver, and may also be referred to as "target", and the sender **110** may be a RDMA sender, and may also be referred to as "initiator".

[0057] Generally, each packet that is sent over the multiplexed connection indicates at least the flow, to which it belongs, and the PSN of the packet. This may be indicated in a header of the packet. Moreover, each message may comprise one or more packets, usually multiple packets. An initial packet of each message indicates at least a total number of packets of that message. This total number may be indicated in a header of the initial packet of the message. The initial packet of the message may indicate additional information regarding the message it belongs to.

[0058] As shown in FIG. **1**, the receiver **100** is configured to receive, from the sender **110**, a first received packet **101**, which is an initial packet of a first message of a first flow. Accordingly, the sender **110** is configured to send, to the receiver **100**, a first sent packet **111**. In the shown case, the first sent packet **111** and the first received packet **101** are the same packet.

[0059] The receiver **100** is further configured to discard the first packet **111**, if the first message cannot be processed by the receiver **100**. For instance, it can happen that the first message cannot be processed temporarily by the receiver. In this case, the receiver **100** may send a NACK packet **103** to the sender **110**. The NACK packet **103** may, however, be also sent by or via another entity. Accordingly, the sender **110** is configured to receive the NACK packet **103**, for instance, from the receiver **100**. The NACK packet **103** received by the sender **110** includes PSN information, which indicates a PSN of an expected packet, which is expected to be received next at the receiver **100**. Further, the NACK packet **103** received by the sender **110** includes an indication that the first message cannot be processed by the receiver **100**.

[0060] The sender **110** is, upon receiving the NACK packet **103**, configured to abort sending any further packet of the first message, since it knows that the first message cannot be processed by the receiver **100**. Further, the sender **110** is configured to send, to the receiver **100**, a second sent packet **112** that does not belong to the first flow, to which the first sent packet **111** and the first message belong, since the first flow may be blocked at the receiver **100**. The sender **110** is configured to send the second sent packet **112** having a PSN that is equal to the sum of the PSN of the first sent packet **111** plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was sent to the receiver after the first sent packet **111**.

[0061] The receiver **100** is configured to receive a second received packet **102**. The second received packet **102** may be the same as the second sent packet **112**, but not necessarily sol. For instance, the sender **110** could have sent other packets after sending the first sent packet **111** and before receiving the NACK packet **103**. In this case, the second received packet **102** may not be the same as the second sent packet **112**. It can also be that packets get lost between the sender **110** and the receiver **100**, so that the second sent packet **112** does not arrive at the receiver **100**, but that another packet arrives at the receiver **100** as the second received packet **101**.

[0062] The receiver **100** may therefore determine whether to discard or not discard the second received packet **102**. The receiver **100** is configured to discard the second received packet **102** (indicated by X) under each one of the following conditions. (1) The second received packet **102** is of the first message. This is, because the first message cannot be processed by the receiver **100**. (2) The second received packet **102** is of a second message and belongs to the first flow. This is, because the first flow may be considered blocked at the receiver **100**. (3) The second received packet **102** is of a third message and does not belong to the first flow, and the PSN of the second received packet **102** is not equal to the sum of the PSN of the first received packet **101** plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet **101**. In this case, notably, the second received packet **102** is not the same as the second sent packet **112** (different PSN), and the second received packet **102** is not the packet that is expected at the receiver **100**.

[0063] The receiver **100** may further maintain flow status information, which indicates one or more blocked flows of the multiple flows. That is, the receiver **100** may maintain information on which flows are blocked. If a flow is blocked, the receiver **100** may discard received packets that belong to this flow. For instance, the receiver **100** may discard the second received packet **102** under the condition that the second received packet **102** belongs to a second flow that is a blocked flow according to the flow status information.

[0064] The receiver **100** may be further configured to not discard the second received packet **102** (as indicated by 0) under the condition that the second received packet **102** is of a third message and does not belong to the first flow or any other blocked flow, and the PSN of the second received packet **102** is equal to the sum of the PSN of the first received packet **101** plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet **101**. For instance, as in FIG. **1** the first received packet **101** is the same as the first sent packet **111**, the second received packet **102** is the same as the second sent packet **112** in this case.

[0065] In the above way, the sender **110** and the receiver **100**, which are configured to operate according to the protocol of sending, receiving, and discarding packets, enable a HOL blocking free transport of packets over the multiplexed connection between the sender and the receiver.

[0066] The sender **110** and the receiver **100** may each comprise a processor or processing circuitry configured to perform, conduct or initiate the various operations of the sender **110** and the receiver **100** described herein. The processing circuitry may comprise hardware and/or the processing circuitry may be controlled by software. The hardware may comprise analog circuitry or digital circuitry, or both analog and digital circuitry. The digital circuitry may comprise components such as application-specific integrated circuits (ASICs), field-programmable arrays (FPGAs), digital signal processors (DSPs), or multi-purpose processors. The sender **110** and the receiver **100** may each further comprise memory circuitry, which stores one or more instruction(s) that can be executed by the processor or by the processing circuitry, in particular under control of the software. For instance, the memory circuitry may comprise a non-transitory storage medium storing executable software code which, when executed by the processor or the processing circuitry, causes the various operations of the sender **110** and the receiver **100** to be performed. In one embodiment, the processing circuitry comprises one or more processors and a non-transitory memory connected to the one or more processors. The non-transitory memory may carry executable program code which, when executed by the one or more processors, causes the sender **110** and the receiver **100** to perform, conduct or initiate the operations or methods described herein.

[0067] This protocol proposed by this disclosure may extend multiplexed connections such as RD connections, to support multiple, as in RC, in-flight messages. Any received message at the receiver **100**, which at least temporarily cannot be processed at the receiver **100**, may be delayed by the NACK message **103**, which may be a receiver not ready (RNR) message. The RNR message affects only the RNR-causing flow at the sender **110**, which allows to avoid HOL blocking. The order of the messages of the RNR-causing flow can be preserved, however, by using the proposed protocol.

[0068] Notably, in case of the sender **110** and receiver **100** being RDMA initiator and RDMA target, such a RNR message can be triggered either by the initial packet of a SEND operation performed by the sender **110**, or by last packet of a WRITE-with-Immediate operation performed by the sender **110**. For example, in InfiniBand (IB) or in RDMA over Converged Ethernet (RoCE), the initial packet of a WRITE-with-Immediate operation does not include any indication that the last packet will carry immediate data that will consume a RQ WQE.

[0069] A shown in FIG. **2**, the receiver **100** according to this disclosure may be further configured to maintain PSN information **201**, which indicates the PSN of an expected packet (ESPN), which is expected to be received next from the sender **110**. Further, each message may be associated with a MSN, and the receiver **100** may be further configured to maintain MSN information **202**, which indicates the MSN of an expected message, which is expected to be processed next. Likewise, the sender **110** may be further configured to maintain SSN information **205**, which indicates a MSN of a next message to be sent to the receiver **100**.

[0070] The PSN information **201** and the SSN information **205** and/or the MSN information **202** may be used for synchronization between the sender **110** and the receiver **100**, The receiver **100** may, for instance, increment the MSN indicated by the MSN information **202** for each message (e.g., WQE) completed (in-order). The receiver **100** may also increase the MSN indicated by the MSN information **202** by one, if the first message cannot be processed. That is, a message of a flow that triggers an RNR message (and leads to a blocking of this flow). and all following messages of the blocked flow may be “counted” as completed by increasing the MSN.

[0071] Further, if the first message cannot be processed by the receiver **100**, the receiver **100** may increase the PSN of the expected packet by the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet **101**. That is the PSN of the expected packet, the EPSN, may be “skipped” according to number of remaining packets of a blocked message or flow. For example, in case of RDMA, if a WRITE-with-Immediate WQE that triggers a RNR message (by its last packet), the EPSN may be updated by +1.

[0072] Notably, for RDMA, a WQE may include a “Started” 1 bit flag to indicate which WQEs were already transmitted and which can be “skipped”. For instance, the 1 bit flag may be ‘1’ when its initial packet is transmitted, and the 1 bit flag may be ‘0’ when a WQE is suspended.

[0073] The receiver **100** may be further configured to send an ACK packet to the sender **100**, if the second received packet **102** is not discarded. The ACK message may include the PSN of the second received packet **102**, the MSN of the message the second received packet **102** belongs to, and the

MSN list comprising, for each blocked flow, the MSN of the first blocked message of said blocked flow. The receiver **100** may also send one or more NACK messages to the sender **110**, wherein any NACK message may include an MSN list comprising. for each blocked flow, the MSN of the first blocked message of said blocked flow. Alternatively, any NACK message may include a PSN list comprising, for each blocked flow, the PSN of the initial packet of the first blocked message of said blocked flow. That is, any ACK or NACK packet (which may also serve as RNR message) may include the PSN, the MSN, and/or a list of “blocked MSNs”.

[0074] In case of RDMA, the sender **110** may retransmit an initial packet of a message (WQE) of a blocked flow only, if the 1 bit flag of the WQE is ‘**1**’ (TRUE). Thereby, the sender **110** may make sure that the receiver **100** updates the EPSN correctly. Otherwise, the WQE may be skipped. Further, the sender **110**, when resuming the WQE that caused the RNR message, may be configured to resume a SEND or resume a WRITE-with-Immediate. In the former case, the entire WQE may be retransmitted and only the initial packet may have a “Resume”=‘1’ indication. In the latter case, only the last packet may be retransmitted, and may include a “Resume”=‘1’indication.

[0075] FIG. **2** illustrates an example according to this disclosure of a message being completed with a previous blocked message. FIG. **2** shows that a first received packet **101** (same as the first sent packet **111**) of a first message from a flow A gets blocked at the receiver **100**. The receiver **100** accordingly sends the NACK packet **103** to the sender **110**. Also another packet **206** of the same message is disregarded at the receiver **100**, as this is one of the conditions mentioned above to disregard a second received packet **102**. Then, a further packet **207** from flow A arrives at the receiver **100**, but is also skipped at the receiver **100** because flow A is currently blocked. That is, the packet **207** is discarded according to one of the conditions mentioned above, under which a second received packet **102** is discarded at the received **100**. A further NACK packet **203** is sent to the sender **110**. Then, a further packet **112** (corresponding to the second sent packet **112** in FIG. **1**) from a flow B is accepted by the receiver **100**. Accordingly, an ACK packet **204** is sent to the sender **110**. The sender **110** may retry the blocked messages (here exemplarily WQE1, WQE2) after a timeout.

[0076] FIG. **3** illustration an example of overcoming packet loss in presence of a blocked flow. FIG. **3** shows that a first received packet **101** of a first message is discarded, wherein the first received packet **101** is the first sent packet **111**. and flow A to which the first message belongs gets blocked. A NACK packet **103** is sent to the sender **110**. Also another packet **301** of the same message is disregarded at the receiver **100**, as this is one of the conditions mentioned above to disregard a second received packet **102**. Then, a further packet **112** of a second message from flow A is lost, wherein this packet could be the second sent packet **112** shown in FIG. **1**. Then, when a further packet **302** of a third message from a flow B arrives at the receiver **100**, it cannot be accepted by the receiver **100**, because its PSN is larger than the EPSN. Thus, it does not correspond to the second sent packet **112**, which is the expected packet at the received **100**. Accordingly, another NACK packet **304** (including the blocked MSN list) is sent back to the sender **110**. When the sender **110** receives the NACK packet **304**, the sender **110** may retransmit the second message although it belongs to a blocked flow A, since it was already transmitted before.

[0077] FIG. **4** shows an example of aborting a message and skipping a new message. FIG. **4** shows particularly that a blocked message can be aborted without having to complete its transmission, and that a message belonging to a blocked flow that was not yet transmitted can be skipped. As shown in FIG. **4**, first, the sender **110** transmits the first three packets **111**, **401**, **402** (out of 20 packets) of a first message, wherein the initial packet **111** correspond to the first received packet **101** at the receiver **100**. Then, the first message is blocked by the receiver **100** when the initial packet **101** is received. The second and third packets **401**, **402** of the first message are considered duplicates by the receiver **100**, because it already adjusted the EPSN to be the PSN of the initial packet of the next message. When the sender **110** receives the RNR message (NACK packet **103**) containing SSN=1 in the list of blocked MSNs, the sender **110** aborts the transmission of the first message, sets the next PSN (NSPN) to be the PSN after the (not transmitted) last packet of the aborted message, and increments its SSN **205**, and marks the flow as blocked. Then, the second message of flow A is skipped, because flow A is blocked and it was not yet transmitted. Then, an initial packet **112** of a third message of flow B is transmitted with the NPSN, and it corresponds with SSN=2, which would have been assigned to the second message, if it would be transmitted before flow A was blocked. This initial packet corresponds to the second sent packet **112** of FIG. **1** and is accepted by the receiver **100**. Accordingly, an ACK packet **403** is sent to the sender **110**.

[0078] In the following, some optional embodiments and alternative implementations are described.

[0079] For example, in case of any different RDMA implementation (iWARP, OmniPath, etc.) is used, then the packet that carries the immediate data for WRITE-with-Immediate operation will trigger the RNR. If SEND-with-Immediate is divided into a separate packet then the packet that carries the immediate data will trigger the RNR and we'll handle it like WRITE-with-Immediate operation.

[0080] Optionally, the retransmission of packets or messages after RNR could be using a no operation (NOP) packet with the number of packets and message number to be skipped. Alternatively, only a header could be sent with no payload of all the missing packets. Alternatively, a single packet header may be used to indicate the packet range either on the header or inside the payload.

[0081] Optionally, if multiple messages are dropped and need to be retransmitted, the above procedure can be used with the indication of how many messages needs to be skipped and not just the number of packets (PSN).

[0082] Optionally, in some scenarios, such as page fault, the RNR can be triggered by any packet of any message. To better handle such scenarios, the receiver **100** may hold and optionally report the PSN of the RNR-causing packet. This may prevent duplicate message execution (if PSN is held by the receiver **100**). This may avoid unnecessary retransmissions (if PSN is reported by the receiver **100** and held by the sender **110**). If the PSN is only held by the receiver **100**, but not reported, the sender **110** might retransmit the entire message. The receiver **100** can calculate the offset of the message to be executed. Packets below the offset can be discarded to prevent duplicate execution. If the PSN is also reported to the sender **110**, the sender **110** may resume the message from the RNR-causing offset.

[0083] Optionally, the solution of this disclosure can work correctly also if the PSN is carried instead of the MSN to report blocked messages, since the PSN can be directly translated into the MSN. However, the MSN is preferred to simplify implementations.

[0084] FIG. **5** shows a method **500** for receiving packets of messages from multiple flows multiplexed onto a connection between a receiver **100** and a sender **110**, according to this disclosure. The method **500** may be performed by the receiver **100**. Each packet indicates at least the flow to which it belongs and its PSN and wherein an initial packet of each message indicates at least a total number of packets of that message.

[0085] The method **500** comprises a step **501** of receiving, from the sender **110**, a first received packet **101** that is an initial packet of a first message of a first flow. The method **500** further comprises a step **502** of discarding the first received packet **101**, if the first message cannot be processed by the receiver **100**. Then, the method **500** comprises a step **503** of receiving, from the sender **110**, a second received packet **102**. The method **500** further comprises a step **504** of discarding the second received packet **102** under each one of the following conditions: the second received packet **102** is of the first message; the second received packet **102** is of a second message and belongs to the first flow; the second received packet **102** is of a third message and does not belong to the first flow, and the PSN of the second received packet **102** is not equal to the sum of the PSN of the first received packet **101** plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet **101**.

[0086] FIG. **6** shows a method **600** for sending packets of messages on multiple flows multiplexed onto a connection between a receiver **100** and a sender **110**, according to this disclosure. The method **600** may be performed by the sender **110**. Each packet indicates at least the flow to which it belongs and its PSN, and an initial packet of each message indicates at least a total number of packets of that message.

[0087] The method **600** comprises a step **601** of sending, to the receiver **100**, a first sent packet **111** that is an initial packet of a first message on a first flow. Then, the method **600** comprises a step **602** of receiving, from the receiver **100**, a NACK packet **103** that includes PSN information which indicates a PSN of an expected packet, which is expected to be received next at the receiver **100**, and an indication that the first message cannot be processed by the receiver **100**. The method **600** further comprises a step **603** of aborting sending any further packet of the first message, and a step **604** of sending, to the receiver **100 100**, a second sent packet **112** that does not belong to the first flow, wherein the PSN of the second sent packet **112** is equal to the sum of the PSN of the first sent packet **111** plus the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was sent to the receiver after the first sent packet **111**.

[0088] The main differences of the solutions of this disclosure to conventional solutions are as follows. RD is limited to single outstanding message per logical connection, and single outstanding message per physical connection. RD uses a special RESYNC opcode to synchronize PSN and MSN between initiator and target after the initiator receives the RNR signal. No data can be sent until RESYNC operation is completed, which may require at least one additional round trip time (RTT). SRD does not maintain order between messages of the same flow, and is limited to single packet message of a limited number of opcodes. DC connects and disconnects before and after switching between flows.

[0089] The present disclosure has been described in conjunction with various embodiments as examples as well as implementations. However, other variations can be understood and effected by those persons skilled in the art and practicing the claimed matter, from the studies of the drawings, this disclosure and the independent claims. In the claims as well as in the description the word “comprising” does not exclude other elements or steps and the indefinite article “a” or “an” does not exclude a plurality. A single element or other unit may fulfill the functions of several entities or items recited in the claims. The mere fact that certain measures are recited in the mutual different dependent claims does not indicate that a combination of these measures cannot be used in an advantageous implementation.

Claims

1. A receiver for receiving packets of messages from multiple flows multiplexed onto a connection between the receiver and a sender, wherein each packet indicates at least a flow to which the each packet belongs and a packet sequence number (PSN), wherein an initial packet of each message indicates at least a total number of packets in that message, and wherein the receiver is configured to: receive, from the sender, a first received packet that is an initial packet of a first message of a first flow; discard the first received packet, if the first message cannot be processed by the receiver; receive, from the sender, a second received packet; and discard the second received packet under each one of the following conditions: the second received packet is of the first message; the second received packet is of a second message and belongs to the first flow; the second received packet is of a third message and does not belong to the first flow, and a PSN of the second received packet is not equal to a sum of a PSN of the first received packet plus a total number of packets of the first message plus a total number of packets of any message that belongs to any flow and was discarded after discarding the first received packet.

2. The receiver according to claim 1, further configured to: maintain flow status information indicating one or more blocked flows of the multiple flows; and discard the second received packet under a condition that the second received packet belongs to a second flow that is a blocked flow according to the flow status information.
3. The receiver according to claim 2, wherein if the first message cannot be processed by the receiver, the receiver is further configured to modify the flow status information to indicate that the first flow is a blocked flow.
4. The receiver according to claim 2, further configured to: maintain PSN information which indicates a PSN of an expected packet, which is expected to be received next from the sender; discard the second received packet under a first condition that the PSN of the second received packet is not equal to the PSN of the expected packet as indicated by the PSN information; and not discard the second received packet under a second condition that the second received packet belongs to a third flow that has a status, according to the flow status information, which indicates that the third flow is not blocked, and if the PSN of the second received packet is equal to the PSN of the expected packet as indicated by the PSN information, and if the third message can be processed by the receiver.
5. The receiver according to claim 4, wherein based on determining that the first message cannot be processed by the receiver, the receiver is further configured to: increase the PSN of the expected packet by the total number of packets of the first message plus the total number of packets of any second message that belongs to the first flow and was discarded after discarding the first received packet.
6. The receiver according to claim 5, wherein each message is associated with a message sequence number (MSN) and the receiver is further configured to: maintain MSN information which indicates an MSN of an expected message, which is expected to be processed next; and increase the MSN indicated by the MSN information by one, if the first message cannot be processed.
7. The receiver according to claim 6, further configured to increase the MSN indicated by the MSN information by one for each second message that belongs to the first flow and was discarded after discarding the first received packet.
8. The receiver according to claim 6, further configured to increase the MSN indicated by the MSN information by one, if the second received packet is a last packet of a message and belongs to the third flow.
9. The receiver according to claim 8, further configured to: send a first not-acknowledge (NACK) packet to the sender, if the first message cannot be processed by the receiver; wherein the first NACK packet includes the PSN information and an indication that the first message cannot be processed by the receiver.
10. The receiver according to claim **49**, further configured to: send a second NACK packet to the sender, if the second received packet is discarded and if the second received packet is of the second message or of the third message; wherein the second NACK packet includes the PSN information and an indication that the first message cannot be processed by the receiver.
11. The receiver according to claim 10, wherein the first NACK packet and the second NACK packet includes the MSN of the first message that cannot be processed by the receiver.
12. The receiver according to claim 11, wherein: the first NACK packet and the second NACK packet includes an MSN list comprising, for each blocked flow, the MSN of the first blocked message of the blocked flow; or the first NACK packet and the second NACK packet includes a PSN list comprising, for each blocked flow, the PSN of the initial packet of the first blocked message of the blocked flow.
13. The receiver according to claim 5, further configured to: send an acknowledge (ACK) packet to the sender, if the second received packet is not discarded; wherein the ACK message includes the PSN of the second received packet, the MSN of the message the second received packet belongs to, and the MSN list comprising, for each blocked flow, the MSN of the first blocked message of the

blocked flow.

14. The receiver according to claim 2, further configured to: receive, from the sender, a third received packet of a fourth message; discard the third received packet in case of a predetermined failure condition and modify the flow status information of the flow to which the fourth message belongs to indicate that this flow is a blocked flow; send a third not-acknowledge (NACK) packet to sender, if the third received packet is discarded; and hold the PSN of the third received packet and/or the MSN of the fourth message.

15. The receiver according to claim 14, wherein the third NACK packet includes the PSN of the third received packet and/or the MSN of the fourth message.

16. The receiver according to claim 1, wherein the connection between the receiver and the sender is a Remote Direct Memory Access (RDMA) connection and/or is a Reliable Datagram (RD) connection.

17. A sender sending packets of messages on multiple flows multiplexed onto a connection between the sender and a receiver wherein each packet indicates at least the flow to which the each packet belongs and a packet sequence number (PSN), and an initial packet of each message indicates at least a total number of packets in that message, and wherein the sender is configured to: send, to the receiver, a first sent packet that is an initial packet of a first message on a first flow; receive, from the receiver, a not-acknowledge (NACK), packet that includes PSN information which indicates a PSN of an expected packet, which is expected to be received next at the receiver, and an indication that the first message cannot be processed by the receiver; abort sending any further packet of the first message; and send, to the receiver, a second sent packet that does not belong to the first flow, wherein a PSN of the second sent packet is equal to a sum of the PSN of the first sent packet plus a total number of packets of the first message plus a total number of packets of any second message that belongs to the first flow and was sent to the receiver after the first sent packet.

18. The sender according to claim 17, wherein the NACK packet includes a message sequence number (MSN) list comprising, for each blocked flow, the MSN of the first message that cannot be processed by the receiver or is discarded by the receiver, and the sender is further configured to: skip sending any first transmission of a packet that belongs to a flow, on which any message that cannot be processed by the receiver is sent.

19. The sender according to claim 17, further configured to: maintain send sequence number (SSN) information which indicates an MSN of a next message to be sent; and send the second sent packet, and associate the second message with a MSN as indicated by the SSN information.

20. The sender according to claim 18, further configured to associate the second message with the MSN of the third message, if the third message was to be sent before the second message but was skipped.

21-23. (canceled)